

Virtual Loupes: An Augmented Reality Aid for Microsurgery

Cory Ilo

Center for Human-Computer Interaction
Virginia Tech

Waylon Zheng

Virginia Tech Carilion School of
Medicine

Doug A. Bowman

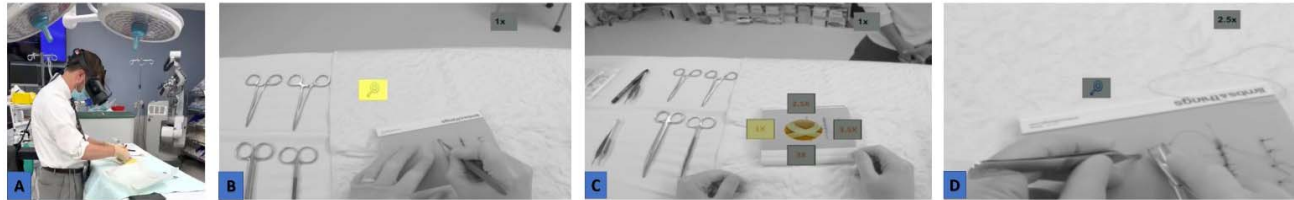
Center for Human-Computer Interaction
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Fig. 1 : A usage example of Virtual loupes: a) A participant performing the study task, b) The default (1x) view inside the headset, c) The discrete zoom menu, d) The zoomed in image after selecting 2.5x magnification.

ABSTRACT

Microsurgery requires the use of loupes, which are optical elements attached to eyeglass frames that magnify objects such as blood vessels and nerves. To address the inflexibility of loupes, we have developed an augmented reality system that provides virtual loupes with a video see-through headset, allowing for easy-to-use variable zoom in a surgical context. Our prototype solution utilizes a gaze-controlled interface, which allows for the zoom level to be selected in either a discrete or continuous manner. A feasibility study revealed both the potential of virtual loupes and the limitations of current technology to realize this concept.

Keywords: Augmented reality, Microsurgery, Gaze-controlled interfaces, Video see-through headset.

Index Terms: •Human-centered computing~Human computer interaction (HCI)~Interaction paradigms~Mixed / augmented reality

1 INTRODUCTION

Microsurgery is a surgical discipline that uses magnification and precision instruments to perform anastomosis of blood vessels and nerves, and other specialized surgical tasks. Microsurgeons have several magnification equipment options to choose from; these include an operating room microscope or physical surgical loupes. Each option has its own set of trade-offs. Surgical microscopes provide magnification levels ranging from 5x to 40x magnification but suffer from poor ergonomics, a single viewing angle, long setup time, and a lack of portability. Further they are expensive compared to surgical loupes.

Surgical loupes are magnifying eyewear typically custom fitted to the wearer. Compared to microscopes, they can reduce the risk of physical discomfort such as eye strain and neck or back pain. However, physical loupes typically offer only one fixed level of magnification. When multiple levels of magnification are needed, the surgeon's loupes must be removed and swapped out, which is a cumbersome process. Higher powered loupes also have heavier weight with increased moment arm, reduced depth of field, and lowered field of view.

In this work, we explore the concept of *Virtual Loupes*, which replace traditional surgical loupes with a video see-through augmented reality (AR) display. The primary advantage of virtual loupes is that the level of magnification can be varied dynamically, or magnification can be turned off altogether.

Prior research provides evidence confirming the feasibility of this project concept. In a series of experiments, Martin-Gonzalez et al. [2] demonstrated that surgeons can efficiently and effectively complete surgical tasks by using head-mounted virtual loupes which leverage video see-through. Other studies explore the application of extended reality (XR) technology in related clinical settings, including facilitating neuronavigation for brain tumor surgery [1], for supporting clinical tasks associated with visualization of medical information, and for pediatric surgical practice. Within their article Diaz et al., [1] discuss surgical use-case considerations which support Martin-Gonzalez et al., [2] conclusions and design decisions. Collectively these studies share important insights on stakeholder needs, surgical task concerns, and prototype design considerations.

2 VIRTUAL LOUPES

Virtual Loupes recreate the experience of physical loupes by leveraging the pixel-level control afforded by video see-through AR headsets. This allows us to manipulate the user's view to simulate optical magnification, similar to the "digital zoom" capabilities of many cameras with fixed lenses. Our current hardware platform consists of a Varjo XR1 headset and a Windows 10 PC equipped with an Intel Core i9-9900k CPU, NVIDIA GeForce RTX 2080Ti graphics card, and 32GB of RAM. We chose to build our AR system on the Varjo XR1 due to its higher camera resolution in comparison to its competitors, native 100Hz eye-tracking, and its ultra-low latency image pipeline (<15ms). The application was built using Unity v2019.4.9f1 and the Varjo Unity Legacy plugin.

In our prototype, shown in Figure 1, the image from the XR1's camera is first cropped based on the currently set level of magnification. Then, the cropped image is placed in a video texture which is applied to a rectangle that fills the user's field of view. This demonstrates hands-free variable zoom that is significantly faster than changing out a pair of loupes or adjusting a microscope, with the ability to not only fit multiple people but display other salient information to users as well. While our prototype demonstrates the concept of virtual loupes, it still has several

limitations. The current state of the Varjo Unity Legacy plugin does not allow us to provide stereo vision, nor are we able to access color imagery from the cameras. The XR1 can do these things natively; however, Unity developers do not have easy access to manipulate the color and stereo imagery. In addition, our virtual loupes prototype is heavy and low-resolution compared to optical loupes, and the device needs to be tethered. We expect all these limitations to disappear as technology improves. At the time of writing, Varjo has already released the XR3 which has better eye-tracking, better displays, LIDAR, native hand-tracking, and most importantly lower weight.

The design of the interaction techniques for changing the level of magnification was a critical part of the prototype design. We considered various input modalities such as voice, gesture, gaze, and physical input devices. Based on an understanding of the microsurgery process gained through discussions with surgeons, we decided to use gaze input, which is hands-free and always available. The use of voice commands could interrupt communication in the operating room and could result in false activations if multiple headsets were being used at once. Hand gestures were avoided because they could potentially interrupt the flow of surgery. We eliminated peripheral devices such as foot pedals or controllers to avoid unnecessary clutter in the operating room.

We designed two gaze-based interaction techniques: *discrete* and *continuous*. The discrete technique, shown in Figure 1c, uses visual menu to select from four discrete levels of magnification. The user first dwells on a magnifying glass icon for ~ 1 sec, which makes the menu appear. The user then dwells on the desired level of magnification for ~1 sec. The continuous mode allows the user to control a virtual “slider” by winking with the right eye to zoom in continuously or with the left eye to zoom out. With this technique, the level of magnification can be set to any level between 1x and 4x. The discrete method, while easy to understand, is a two-step action with only a limited number of options. The continuous method effectively has an infinite number of options while also being quicker. However, some people are not physically able to sustain a wink with each eye individually.

3 FEASIBILITY STUDY

The primary goal of our user study was to understand the feasibility and potential usefulness of the Virtual Loupes concept in microsurgery while also investigating the technical capabilities of state-of-the-art AR HMDs in this context. In addition, we also sought to gain a preliminary understanding of the usability of the two gaze-based interfaces for controlling zoom level. This led us to the following research questions:

- What is the feasibility and potential usefulness of virtual loupes in microsurgery compared to physical loupes?
- How will the limitations of current state-of-the-art AR HMDs affect the perceived usefulness of virtual loupes?
- What are the relative benefits of discrete and continuous zoom in our AR system?

We ran our study on a total of six participants, all of whom were male surgeons or surgical residents of various experience levels. Most of the participants had no previous experience with AR/VR technology. Each participant was introduced to the concept of virtual loupes and the purpose of the study. Afterwards, the headset was calibrated to each participant to ensure proper operation of the prototype. The procedure consisted of a brief training session and two surgical tasks which involved the suturing of simulated skin at

various magnifications using each gaze-based interface mode. Finally, we had the participant complete a post-study survey consisting of system usability scale (SUS) [4] questions and an informal interview.

4 RESULTS & DISCUSSION

All participants except one were able to complete the suturing tasks using the virtual loupes. However, the results from the SUS survey indicate poor usability. Most participants felt the device impeded their ability to complete the surgical tasks; in addition to increasing their general task completion time compared to physical loupes. However, the vast majority found the gaze-based interfaces easy to use.

The findings gathered from the interview provides much needed context and illustrates the potential of this concept. The participants reported many ergonomic issues with the prototype, such as lack of stereo vision, bulky and heavy device, poor resolution when zoomed in, the inability to see in color, and the inability to “look down with your eyes instead your of head.” However, most of the surgeons also recognized the potential benefits and usefulness of Virtual Loupes.

It is important to note that the limitations of the prototype were all technical deficiencies associated with the hardware/software platform. Therefore, while not currently feasible technically, virtual loupes do have the potential to one day be extremely useful in an operative setting. One participant said that we were essentially “combining the benefits of loupes with the benefits of a microscope,” which suggests that it is useful to have variable magnification in a form that is more portable and flexible than a large, fixed microscope.

In comparing the two interaction techniques, participants consistently rated the discrete gaze-based interface mode as easier to use, although both techniques were rated highly. The discrete technique was fast and accurate, while it was sometimes difficult to set the continuous technique to the exact desired level of magnification. At least for the microsurgery tasks considered by our participants, the choice of four discrete magnification levels was sufficient, so the main advantage of the continuous technique (access to any magnification level) was not seen as useful.

Overall, the results of the feasibility study demonstrated that while it is possible today to use virtual loupes to complete microsurgical tasks, significant technical advances are still needed to make them usable, efficient, and comfortable. The gaze-based interaction techniques were validated as easy to use and appropriate for the surgical context.

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